Conditional Statements (if / else if / else)

Question 1: Grade Classifier

Topic	if / else if / else

Write a C++ program that reads a student's exam score (0–100) and prints the corresponding grade:

- 90 – 100 → A
- 80 – 89 → B
- 70 – 79 → C
- 60 – 69 → D
- Below 60 → F (Fail)

If the score is outside the range 0–100, display an error message.

Question 2: Leap Year Checker

Topic	Nested if / logical operators

Write a C++ program that reads a year and determines if it is a leap year.

A year is a leap year if:

- It is divisible by 4 AND not divisible by 100, OR
- It is divisible by 400.

Display whether the year is a leap year or not.

Question 3: Simple Calculator

Topic	if / else if with switch-style logic

Write a C++ program that reads two numbers and an operator (+, -, *, /) from the user, then calculates and displays the result.

Use if / else if statements to handle each operator. Handle division by zero as a special case.

SECTION B: For Loop

Question 4: Multiplication Table

Topic	for loop

Write a C++ program that reads an integer n from the user and prints its multiplication table from 1 to 10.

Expected output format (example for n = 5):

```
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50
```

Question 5: Sum of Digits

Topic	for loop with arithmetic

Write a C++ program that reads a positive integer and calculates the sum of all its digits.

Example: Input 1234 → Output: Sum of digits = 10

Hint: Use the modulus operator (%) to extract digits and integer division (/) to remove them.

Question 6: Pattern Printing

Topic	Nested for loops

Write a C++ program that reads an integer n and prints the following right-angled triangle pattern.

Example for n = 5:

```
*
* *
* * *
* * * *
* * * * *
```

SECTION C: While Loop

Question 7: **Number Guessing Game** [10 marks]

Topic	while loop with conditions

Write a C++ program that implements a simple number guessing game.

The secret number is fixed at 42 (hardcoded).

The program should:

1. Keep asking the user to guess until they get the correct answer.
2. Print 'Too high!' if the guess is greater than 42.
3. Print 'Too low!' if the guess is less than 42.
4. Print 'Correct! You guessed it in X tries.' when correct.

— End of Exercise Sheet —